

# MUHAMMAD SIDDIQ

## Generative AI Engineer

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### PROFILE SUMMARY

GenAI & Backend Engineer with experience building production-grade AI applications, RAG systems, Agentic AI workflows, and scalable backend services. Skilled in developing intelligent solutions using Python, LangChain, LangGraph, FastAPI, Flask, OpenAI APIs, and vector databases to deliver context-aware and automation-driven experiences. Proven ability to design, develop, and deploy end-to-end products ranging from AI chatbots and WhatsApp agents to SaaS platforms and enterprise web applications. Passionate about transforming business requirements into reliable, maintainable, and impactful AI-powered products while bridging modern AI capabilities with real-world software engineering.

### TECHNICAL SKILLS

|                          |                                                                                                                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages:               | Python (Proficient)                                                                                                                                   |
| Artificial Intelligence: | Machine Learning, Deep Learning, NLP, LLM, Agentic AI, RAG Pipelines, Generative AI, AI Automation, Voice-to-Voice AI Agents, MCP                     |
| Frameworks & Libs:       | LangChain, LangGraph, LangSmith, FastAPI, Flask, React, Next.js, Node.js, NumPy, Pandas, Scikit-learn, TensorFlow/Keras, Pydantic, SQLAlchemy, Prisma |
| Databases & Vector:      | MySQL, PostgreSQL, Redis (Caching), Supabase, PostgreSQL, Pinecone, Faiss, Chroma                                                                     |
| Tools & Platforms:       | Hugging Face, Postman, Twilio, MCP                                                                                                                    |
| Deployment & VC:         | Vercel, Netlify, Railway, Hostinger, cPanal, SiteGround, DigitalOcean, , Git, GitHub                                                                  |
| Software :               | VS Code, Jupyter Notebook, Google Colab                                                                                                               |
| Soft Skills:             | Adaptability, Time Management, Quick Learning, Effective Multitasking                                                                                 |
| AI Engineering Tools:    | Cursor, GitHub Copilot, Claude Code, OpenCode, Kilo Code, Hermes Agent, Qwen CLI, Gemini CLI                                                          |

### PROFESSIONAL WORK EXPERIENCE

- GenAI and backend Engineer - ElevateXCrew**Faislabad, Pakistan | Feb 2026 – Present
- Designed and deployed Agentic AI pipelines, RAG systems, AI chatbots, and n8n automation workflows for client projects, reducing manual operations and enabling intelligent business automation.
  - Built Voice-to-Voice AI Agents with STT/TTS integration, an AI-powered Appointment Booking Chatbot, and a Client Project Monitoring System as part of the company's internal product development.

- Designed and delivered hands on curricula on Generative AI, prompt engineering, and LLM orchestration frameworks, enabling learners to build production ready AI applications.
- Mentored students and developers through real world implementation of RAG pipelines, API integrations and vector databases, accelerating technical adoption and practical skills.
- Supervised capstone projects focused on autonomous agents and workflow automation, guiding cohorts from foundational AI concepts to complete end to end system deployments.

## AI/ML Engineer — DevelopersHub Corporation

Remote | Dec 2024 – Jan 2025

- Designed and implemented LLM/GPT based RAG systems and chatbots to enhance language understanding, automate workflows, and support customer knowledge retrieval.
- Developed AI-driven automation solutions, including social media auto-posting and n8n workflows, reducing manual effort and improving operational efficiency.

## CERTIFICATIONS

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**Akhuwat First faislabad** — Backend Engineer

Aug 2025

**Tecrix** — GenAI Engineer

OCT 2025

## PROJECTS

### OpenGPT

*LangGraph, LangChain, FastAPI, React, FAISS, SQLite, OpenAI API, RAG, Vector Embeddings*

- Built an intelligent AI chatbot using LangGraph orchestration, FastAPI backend, and React frontend with real-time streaming responses and multi-thread conversation management.
- Implemented RAG pipeline with FAISS vector database, automated PDF processing, semantic search, and context-aware document retrieval.
- Integrated multi-tool capabilities including web search, weather, calculator, and stock market data with persistent chat history and memory management.

### OpenWA (Open Source Contribution)

*TypeScript, Node.js, Webhooks, WhatsApp Web*

- Added support for downloading documents received through WhatsApp webhooks, enabling AI-powered document ingestion workflows.
- Developed webhook enhancements to capture, retrieve, and persist incoming PDF and DOCX files.
- Improved OpenWA's extensibility for enterprise chatbot and RAG-based automation systems.

### WhatsApp Booking Agent

*OpenAI API, Twilio API, Flask, React, MySQL, Redis, OCR, WhatsApp Business API*

- Developed an AI-powered WhatsApp booking assistant capable of automating appointment scheduling through natural language conversations.
- Built a full-stack platform with real-time booking management, analytics dashboard, automated reminders, and Redis-powered background processing.
- Implemented OCR-based payment verification, structured booking workflows, and AI-driven conversation handling to improve booking accuracy and customer experience.

**YouTube RAG Chat – Chrome Extension**

*FastAPI, Chrome Extension, FAISS, OpenAI API, YouTube Transcript API, RAG, Vector Embeddings*

- Built a Chrome extension that enables users to chat with YouTube videos using transcript extraction, semantic search, and AI-powered question answering.
- Developed a FastAPI backend for transcript processing, vector embedding generation, and context-aware retrieval using FAISS.
- Implemented conversation persistence, automatic video detection, and real-time retrieval workflows for instant video insights.

**LMS Agentic**

*LangGraph, OpenAI API, React, TypeScript, Flask, SQLAlchemy, PostgreSQL/MySQL, Agentic AI*

- Developed a full-stack Learning Management System with role-based access for administrators, teachers, and students.
- Built an AI-powered educational assistant using LangGraph and OpenAI API with role-specific tools and workflow automation.
- Implemented agent orchestration, persistent chat history, and comprehensive course management features through a unified dashboard.

**ElevateXCrew – Corporate Website & CMS Platform**

*Next.js 14, React, TypeScript, Tailwind CSS, Node.js, Express.js, Prisma ORM, MySQL, JWT, Vite, Axios, Multer, cPanel, SSR,*

- Led the end-to-end development lifecycle, transforming business requirements into a production-ready corporate website and content management platform for a software development company.
- Architected and implemented a full-stack solution with Next.js SSR frontend, Express.js backend, MySQL database, and dedicated React admin portal supporting blogs, portfolios, careers, and media management.
- Delivered SEO-optimized, scalable, and maintainable infrastructure featuring JWT authentication, markdown-based CMS, image management, dynamic content rendering, automated cache revalidation, and production deployment.

**RaketH Clone – AI Voice Cloning & Text-to-Speech SaaS Platform**

*Next.js 16, TypeScript, Prisma ORM, NextAuth.js, MySQL, Tailwind CSS, Zustand, TanStack Query, Framer Motion, Voice Cloning AI, Text-to-Speech (TTS), Translation AI, SaaS Architecture*

- Designed and developed an end-to-end AI SaaS platform enabling voice cloning, multilingual speech synthesis, and translation-powered voice generation through a scalable subscription-based system.
- Engineered a production-ready full-stack application with secure authentication, credit-based billing, generation history, voice library management, and responsive user dashboards.
- Built administrative tools for real-time platform analytics, user and subscription management, credit allocation, and customer engagement workflows, supporting commercial deployment of AI voice services.

**GraceWise – AI-Powered Spiritual Learning Platform**

*Flask, LangChain, OpenAI API, Groq, ChromaDB, Sentence Transformers, MySQL, SQLAlchemy, RAG, Semantic Search*

- Engineered the Curriculum Management system for creating, organizing, and delivering structured educational content across multiple age groups and learning stages.
- Built a production-ready RAG chatbot leveraging PDF knowledge bases, vector search, and LLM-powered reasoning to provide personalized spiritual guidance.
- Developed end-to-end document processing pipelines including PDF extraction, embedding generation, vector storage, and retrieval-augmented response generation.

**EDUCATION**

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|-------------------------------------------------------------|-------------------------------|
| <b>Intermediate – Akhuwat College Kasur</b>                 | Kasur, Pakistan   2022 – 2024 |
| <b>BS Computer Science – Virtual University of Pakistan</b> | 2024 – 2028                   |